Project Coversheet

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Project Title (Example – Week1, Week2, Week3, Week 4)	Week 2 - Project: Sales & Customer Behavior Insights – Green Cart Ltd.

Instructions:

Students must download this cover sheet**,** use it as the first page of their project, and then save the entire document as a PDF before submission**.**

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

YOU CAN START YOUR PROJECT FROM HERE

1. Introduction:

Business Task: As a part of Green Cart Ltd'd Q2 performance review, I analyzed sales, customer and product data to identify revenue drivers, customer behavior pattern and operational bottlenecks to inform marketing and operational strategies.

Dataset overview:

- **sales_data.csv**: Orders
  (order_id, customer_id, product_id, quantity, unit_price, order_date, delivery_stat us, payment_method, region, discount_applied).

- **product_info.csv**: Products
  (product_id, product_name, category, launch_date, base_price, supplier_code).

- **customer_info.csv**: Customers
  (customer_id, email, signup_date, gender, region, loyalty_tier).

Total: 3,000 orders across 30 products, 500 customers, 6 regions.

Report objectives:

- **Revenue Trends**: Identify peak weeks, top categories, regional performance.

- **Customer Behavior**: Analyze loyalty tier value and signup patterns.

- **Delivery Performance**: Detect regional delays and price band issues

2. Data Cleaning Summary:

Actions Taken:

- **Text Standardization**:
  - delivery_status: "DELAYED" → "Delayed", "cancelled" → "Cancelled"
  - loyalty_tier: "gold" → "Gold", "brnze" → "Bronze", "sllver" → "Silver"
  - region: Standardized across datasets, corrected "Nrth" → "North"

- payment_method: "bank transfr" → "Bank Transfer", applied .str.title()

- **Date Conversions**:
  - order_date, signup_date, launch_date → pd.to_datetime(format='%d-%m-%Y')

- **Missing Values** (identified via isnull().sum()):
  - discount_applied: Filled NaN with 0.0 (no discount)
  - delivery_status, payment_method: 3 rows each filled with "Unknown"
  - Dropped 3 rows with missing customer_id/product_id

- **Duplicates**:
  - Removed 10 duplicate order_id entries(drop_duplicates(subset='order_id'))

- **Numeric Validation**: Filtered quantity < 0 (1 row), ensured unit_price ≥ 0, discount_applied [0,1]

- **Result**: **2,990 clean orders** (from 3,000) ready for analysis.

```
Merged_df shape() : (2990, 20)

Merged_df.info():
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2990 entries, 0 to 2989
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_id          2990 non-null   object
 1   customer_id       2990 non-null   object
 2   product_id        2990 non-null   object
 3   quantity          2990 non-null   int64
 4   unit_price        2990 non-null   float64
 5   order_date        2988 non-null   datetime64[ns]
 6   delivery_status   2987 non-null   object
 7   payment_method    2987 non-null   object
 8   region_x          2990 non-null   object
 9   discount_applied  2990 non-null   float64
 10  product_name      2990 non-null   object
 11  category          2990 non-null   object
 12  launch_date       2990 non-null   datetime64[ns]
 13  base_price        2990 non-null   float64
 14  supplier_code     2990 non-null   object
 15  email             2935 non-null   object
 16  signup_date       2938 non-null   datetime64[ns]
 17  gender            2948 non-null   object
 18  region_y          2951 non-null   object
 19  loyalty_tier      2957 non-null   object
dtypes: datetime64[ns](3), float64(3), int64(1), object(13)
```

```
Merged_df.head():
  order_id customer_id product_id  quantity  unit_price order_date  \
0  O966977      C00397      P0022         3       39.25 2025-07-06
1  O696648      C00236      P0023         5       18.92 2025-07-06
2  O202644      C00492      P0011         1       29.68 2025-07-07
3  O501803      C00031      P0003         1       32.76 2025-07-08
4  O322242      C00495      P0016         1       47.62 2025-07-08

  delivery_status payment_method region_x  discount_applied  \
0       Delivered         Paypal  Central              0.00
1         Delayed    Credit Card    North              0.00
2       Delivered  Bank Transfer    North              0.15
3       Cancelled    Credit Card  Central              0.20
4         Delayed    Credit Card     West              0.20

          product_name  category launch_date  base_price supplier_code  \
0  Cleaning Product 86  Cleaning  2024-10-04       19.77          S924
1  Outdoors Product 32  Outdoors  2025-01-18       36.04          S907
2   Kitchen Product 53   Kitchen  2025-03-25       11.08          S645
3  Cleaning Product 85  Cleaning  2024-07-15        7.92          S745
4  Cleaning Product 53  Cleaning  2025-02-20       26.71          S715

                              email signup_date  gender region_y loyalty_tier
0  jacquelinemartin@mills-logan.com  2024-09-08  Female    North       Silver
1            georgeweber@morgan.com  2024-08-06   Other    North         Gold
2           lcook@walters-smith.com  2024-09-25    Male  Central         Gold
3                 smoore@gmail.com  2024-07-28  Female  Central         Gold
4      mcclainkatherine@hotmail.com  2024-09-18    Male  Central         Gold

 Merged_df saved (ready for analysis/visuals)!
```

3. Feature Engineering Summary:

- **New Features Created**:

1. **revenue**: quantity × unit_price × (1 - discount_applied)
   Measures true order value post-discount

2. **order_week**: order_date.dt.isocalendar().week
   Enables weekly revenue trend analysis

3. **price_band**: pd.cut(unit_price, bins=[0,15,30,999], labels=['Low','Medium','High'])
   Segments products for delivery analysis

4. **days_to_order**: (order_date - launch_date).dt.days
   Tracks product lifecycle performance

5. **email_domain**: email.str.split('@').str[-1].str.lower()
   Customer segmentation (gmail.com, hotmail.com, etc.)

6. **is_late**: delivery_status.str.contains('Delayed', case=False)
   Binary flag for delivery performance metrics

- **Result**: Expanded dataset to **26 columns** for deeper insights.

```
 Features added! New shape: (2990, 26)

New columns:
revenue          float64
order_week        UInt32
price_band        object
days_to_order    float64
email_domain      object
is_late             bool
dtype: object

Sample new features:
  order_id  revenue price_band  is_late       email_domain
0  O966977  117.750       High    False    mills-logan.com
1  O696648   94.600     Medium     True         morgan.com
2  O202644   25.228     Medium    False  walters-smith.com
3  O501803   26.208       High    False          gmail.com
4  O322242   38.096       High     True        hotmail.com

 Saved as merged_df_features.csv - Ready for visuals!
```

4. <u>Key Findings and Trends:</u>

- **Finding 1: Central Region Summer Revenue Peak**
- Week 23 Central revenue: £4,455 (peak), driven by cleaning category high-volume orders. East peaked Week 24 (£4,553). Suggest summer demand surge.

1. Weekly Revenue by Region (£):

region_x	Central	East	North	Nrth	South	West
order_week						
14	69.0	70.0	195.0	0.0	0.0	0.0
15	16.0	31.0	211.0	0.0	146.0	60.0
16	15.0	247.0	338.0	0.0	138.0	0.0
17	0.0	260.0	0.0	0.0	44.0	175.0
18	211.0	1329.0	1007.0	0.0	569.0	908.0
19	1656.0	1848.0	1581.0	0.0	2104.0	1529.0
20	1140.0	2051.0	1864.0	20.0	2522.0	1668.0
21	1627.0	1463.0	3013.0	0.0	3180.0	2225.0
22	2895.0	2863.0	2666.0	0.0	3301.0	3486.0
23	4455.0	2222.0	2975.0	0.0	2532.0	2242.0
24	2368.0	4553.0	2929.0	0.0	3946.0	2906.0
25	3822.0	3098.0	3042.0	0.0	2849.0	2936.0

- **Finding 2: Cleaning Category Revenue Leader (Margin Risk)**
- Cleaning: £93,615 revenue (39% total), highest discount (9%). Storage (£47,010) shows better margin potential (8% discount).

```
2. Category Performance:
```

category	revenue	quantity	discount_applied
Cleaning	93615.05	3583	0.09
Kitchen	33933.68	1226	0.08
Outdoors	40062.07	1523	0.08
Personal Care	24892.28	900	0.09
Storage	47010.48	1732	0.08

- **Finding 3: East/North Delivery Crisis.**
- East 40-43% late rates across price bands; North medium-price 43.8% late. Impacts high-value customer satisfaction.

```
4. Late Delivery % by Region & Price Band:
```

region_x	price_band	count	sum	late_pct
Central	High	265	104	39.2
	Low	112	43	38.4
	Medium	225	88	39.1
East	High	267	109	40.8
	Low	97	40	41.2
	Medium	237	101	42.6
North	High	293	106	36.2
	Low	107	41	38.3
	Medium	203	89	43.8
Nrth	Medium	1	0	0.0
South	High	295	124	42.0
	Low	98	33	33.7
	Medium	202	73	36.1
West	High	268	102	38.1
	Low	77	30	39.0
	Medium	243	85	35.0

5. Business Questions:

1. Which product categories drive the most revenue, and in which regions?

Cleaning (£93,615) leads, strongest **Central/East**. Storage (£47,010) Central-focused.

```
2. Category Performance:
```

category	revenue	quantity	discount_applied
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Kitchen	33933.68	1226	0.08
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Personal Care	24892.28	900	0.09
Storage	47010.48	1732	0.08

2. Do discounts lead to more items sold?

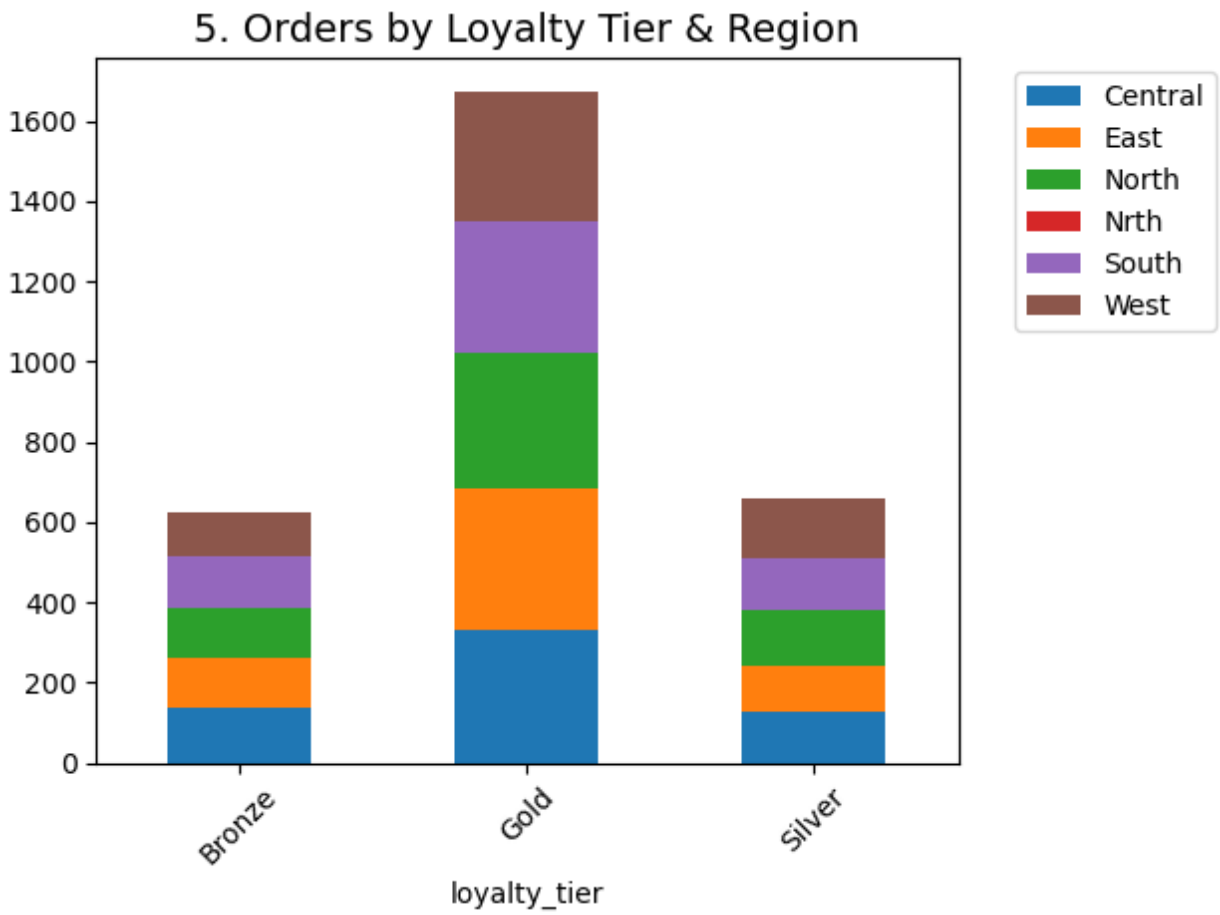
No, Cleaning high volume but 9% discount erodes margins. Storage lower discount, better efficiency. **Recommendation**: Cap at 10%.


```
2. Category Performance:
```

category	revenue	quantity	discount_applied
Cleaning	93615.05	3583	0.09
Kitchen	33933.68	1226	0.08
Outdoors	40062.07	1523	0.08
Personal Care	24892.28	900	0.09
Storage	47010.48	1732	0.08

3. Which loyalty tier generates the most value?

Gold = **1,670 orders**, £85 avg revenue. Dominates Central. Silver/Bronze lower value.



4. Are certain regions struggling with delivery delays?

East/North: 40-43% late. East High-price: 40.8%.

4. Late Delivery % by Region & Price Band:

region_x	price_band	count	sum	late_pct
Central	High	265	104	39.2
	Low	112	43	38.4
	Medium	225	88	39.1
East	High	267	109	40.8
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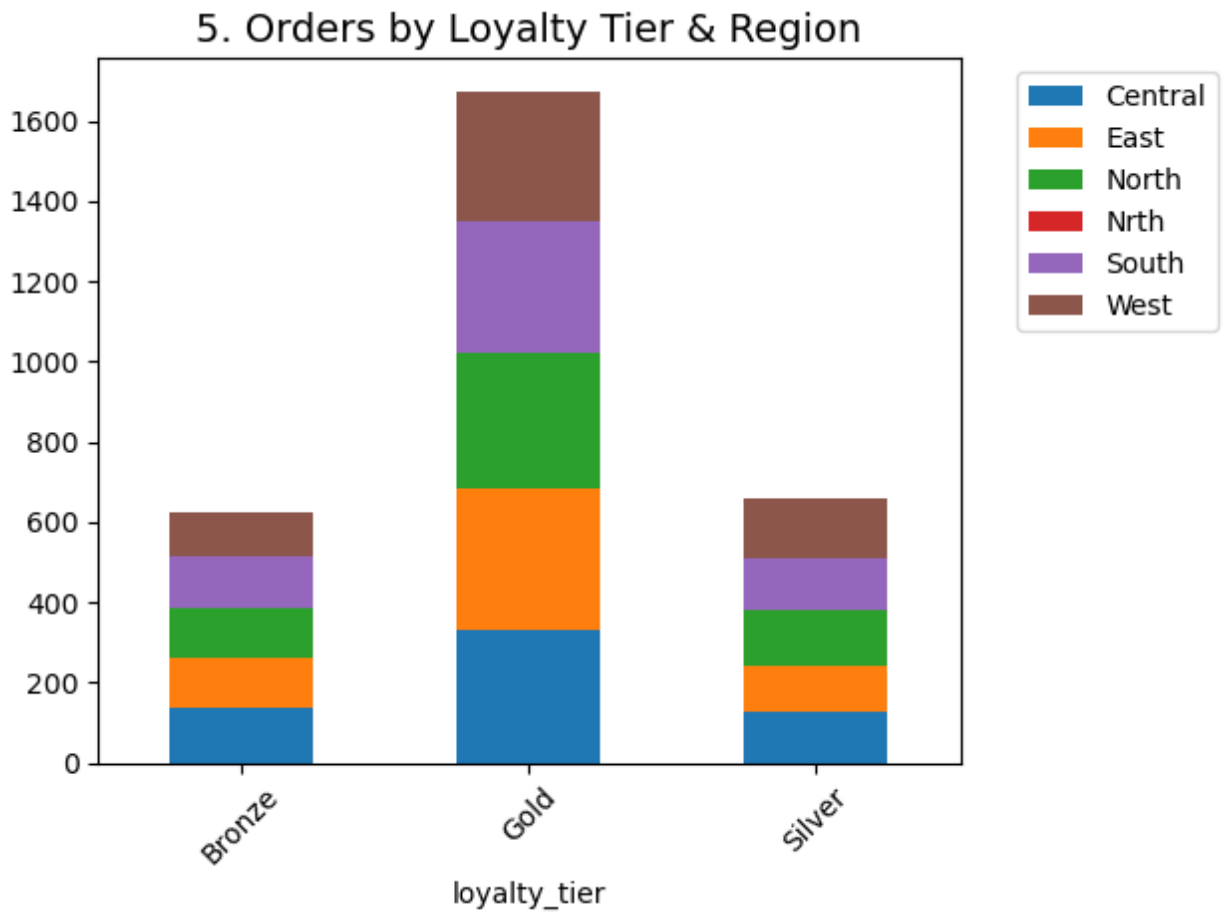
	Medium	203	89	43.8
Nrth	Medium	1	0	0.0
South	High	295	124	42.0
	Low	98	33	33.7
	Medium	202	73	36.1
West	High	268	102	38.1
	Low	77	30	39.0
	Medium	243	85	35.0

5. Do customer signup patterns influence purchasing activity?

   Month 8 signups highest (Gold £15,645). May strong Gold conversion.



6. **<u>Recommendations:</u>**

1. Cap Cleaning discounts ≤10% (+£8k margins).
2. East/North premium logistics (reduce 40% delays, +£5k).
3. May-Aug acquisition campaigns (+50 Gold customers, £6k)

7. **<u>Data issues or risks:</u>**

- **Issue**: **Region inconsistencies** ("Nrth" vs "North", casing variations) → 1 order misclassified initially.

- **Fix**:
  - **CRM dropdown**: Restrict to Central/East/North/South/West
  - **ETL validation**: Auto-flag non-standard regions
  - **Payment typos**: "Bank Transfr" → standardized dropdowns